In [ ]:

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from PyQt5 import QtWidgets

class CalculatorPyQt5(QtWidgets.QWidget):
    def __init__(self):
        super().__init__()
        self.initUI()

    def initUI(self):
        self.setWindowTitle("Calculator Program")
        self.setGeometry(10, 10, 30, 20)

        grid = QtWidgets.QGridLayout()
        self.setLayout(grid)

        grid.addWidget(QtWidgets.QLabel('Number 1'), 0, 0)
        self.num1_entry = QtWidgets.QLineEdit(self)
        grid.addWidget(self.num1_entry, 0, 1)

        grid.addWidget(QtWidgets.QLabel('Number 2'), 1, 0)
        self.num2_entry = QtWidgets.QLineEdit(self)
        grid.addWidget(self.num2_entry, 1, 1)

        # Result label
        self.result_label = QtWidgets.QLabel("Result: -")
        grid.addWidget(self.result_label, 2, 1, 1, 2)

        # Buttons for Addition and Subtraction
        add_button = QtWidgets.QPushButton("Addition", self)
        add_button.clicked.connect(lambda: self.handle_operation('+'))
        grid.addWidget(add_button, 3, 0)

        sub_button = QtWidgets.QPushButton("Subtraction", self)
        sub_button.clicked.connect(lambda: self.handle_operation('-'))
        grid.addWidget(sub_button, 3, 1)

        # Multiplication button
        mul_button = QtWidgets.QPushButton("Multiplication", self)
        mul_button.clicked.connect(lambda: self.handle_operation('*'))
        grid.addWidget(mul_button, 4, 0)

        # Division button
        div_button = QtWidgets.QPushButton("Division", self)
        div_button.clicked.connect(lambda: self.handle_operation('/'))
        grid.addWidget(div_button, 4, 1)

    def handle_operation(self, operation):
        num1 = int(self.num1_entry.text())
        num2 = int(self.num2_entry.text())

        if operation == '+':
            result = num1 + num2
        elif operation == '-':
            result = num1 - num2
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        elif operation == '*':
            result = num1 * num2
        elif operation == '/':
            if num2 == 0:
                self.result_label.setText("Cannot divide by zero")
                return
            result = num1 / num2

        self.result_label.setText(f"Result: {result}")

# Part 2 Testing
if __name__ == "__main__":
    import sys
    app = QtWidgets.QApplication(sys.argv)
    calc_kernel = CalculatorPyQt5()
    calc_kernel.show()
    sys.exit(app.exec_())
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